



Uncertainty evaluation procedure and intercomparison of bell provers as a calibration system for gas flow meters

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ABSTRACT

Many NMIs (national measurement institutes) and calibration laboratories worldwide use a bell prover as a calibration system for gas flow meters. The basic definition and procedure to estimate the bell prover uncertainty have been given in previous studies. After the announcement of a mutual recognition agreement (MRA) in 1998 by the BIPM (Bureau International des Poids et Mesures), many NMIs have needed to have more details of uncertainty evaluation procedure of the bell prover. In this study, more details of the test procedure of bell prover uncertainty are presented with a modification of position of the temperature and the pressure sensor to measure more exact values. The other modification is to use three precise guide rods with bearings to make the pressure in the bell more stable. Furthermore, a laser interferometer is used to enhance the measurement accuracy of the testing time and the traveling length of the bell. The uncertainty of the bells estimated by the procedure given in this study is 0.13% at the confidence level of 95%, rather than 0.17% of the previous one. Recently, the results of CCM.FF-K6, which is an international comparison of NMIs organized by the CIPM (Comité International des Poids et Mesures), show that the uncertainty of the KRISS (Korea Research Institute of Standards and Science) bell prover estimated by this study is reasonable.

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1. Introduction

Recently, an MRA (mutual recognition agreement) of testing and calibration recommended by the BIPM (Bureau International des Poids et Mesures) has been accepted by many NMIs (national measurement institutes) of the world. According to the basic agreement of the MRA, their quality of testing and calibration has to comply with the ISO/IEC 17025 standard [1]. The basic checking point for the quality of measurement results of an accredited calibration body is uncertainty of measurement. The uncertainty of the measurement is a value to express the reliability of the experimental data. Hence it is essential to declare the uncertainty values of the transfer standards of international intercomparisons or a test to have accreditation from international test institutes.

Improvement of measurement accuracy for gas flow rate is very important in fair commercial trading of gases and energy savings. The flow rate of natural gas must be measured in a wide pressure range up to ten times atmospheric pressure. Variations of the gas flow condition change the flow rate from the reference value. So it is essential to calibrate gas flow meters against with a gas flow standard system to ensure accurate flow rate measurement.

The conventional gas flow standard system is a bell or a piston prover. The standard systems must have traceability and official certifications in accordance with the national standards in order to maintain measurement accuracy and reliability.

Bell provers are used as primary standards for the measurement of volumetric gas flow rate in many NMIs. These devices measure the displaced volume and air collection time to calculate the flow rate and, over the years, each NMI has developed its own measurement methods for volume and time in the bell prover.

The bell of the bell prover is an elaborate cylinder. Its top is closed and its open bottom is in oil. In other words, the bell-shaped cylinder rises up when gas flows inside it. As much gas will be captured as the volume equivalent to the distance travelled by the bell. Currently, the distance travelled is measured from outside the bell with a laser interferometer or encoder and the distance travelled is multiplied by the corrected area of the bell to measure the volume. A counterweight is attached to the bell prover as a counterbalance to control the pressure inside the cylinder. The counterweight is designed in such a way that the pressure is constant even if the bell goes up and down. Also, oil is filled the outer prover tank to maintain air tightness and to ensure that the bell can move up and down.

The bell provers at the KRISS (Korea Research Institute of Standards and Science), with an uncertainty 0.17% at 95% confidence level, were about 30 years old. So the repeatability of time and stroke distance measurement of the bell movement was not good

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Nomenclature

C_d	Discharge coefficient
C_{dm}	Averaged discharge coefficient
c_i	Sensitivity coefficient
D_{in}	Bell inside diameter (m)
D_{out}	Bell outside diameter (m)
d	Sonic nozzle throat diameter (m)
$\vec{dA}$	Unit vector of control surface
g	Acceleration due to gravity (m/s^2)
K	Temperature (K)
k	Coverage factor
L	Stroke of bell (m)
m_b	Air mass in the storage volume at the beginning of calibration (kg)
m_c	Collected air mass (kg)
m_e	Air mass in the control and storage volume at the end of calibration (kg)
m_ℓ	Mass of leakage and absorption during the calibration process (kg)
P_b	Pressure in the bell at the beginning of calibration (Pa)
P_e	Pressure in the bell at the end of calibration (Pa)
q_k	k th measurement value
q_m	Mass flow rate (kg/h)
R	Gas constant inside the collecting volume (J/K mol)
T_b	Temperature in the bell at the beginning of calibration (K)
T_e	Temperature in the bell at the end of calibration (K)
t	Collection time (s)
t_m	Timer (s)
t_{BR}	Timing error by the bell rocking (s)
t_R	Resolution of the timer (s)
t_V	Timing error of valve (s)
u_A	Type A standard uncertainty
u_B	Type B standard uncertainty
u_C	Combined standard uncertainty
U	Expanded uncertainty
V	Control volume (m^3)
V_c	Collection volume (m^3)
V_L	Storage volume (m^3)
$V_{oil,\ell}$	Displaced volume outside the bell, due to the change in oil level by the pressure change inside of the bell (m^3)
$V_{oil,ad}$	Volume of oil adhering to the bell (m^3)
V_{out}	Volume outside of the bell (m^3)
v_B	Bell rise velocity (m/s)
$\vec{v}$	Fluid velocity through the control surface enclosing the control volume (m/s)
Z	Gas compressibility constant (J/K mol)
Δm_L	$V_L \times \Delta \rho_{eb}$ (kg)
$\Delta \rho_{eb}$	$\rho_e - \rho_b$ (kg/m^3)
k	Coverage factor
ν	Kinematics viscosity of the sealing oil (m^2/s)
ν_{eff}	Number of effective degrees of freedom
ρ	Air density in the control volume (kg/m^3)
ρ_b	Air density in the bell at the beginning of calibration (kg/m^3)
ρ_e	Air density in the bell at the end of calibration (kg/m^3)

enough for a national standard system. Another problem was that a temperature and a pressure sensor at the inlet pipe on the bell prover had poor measurement quality and needed to be moved to

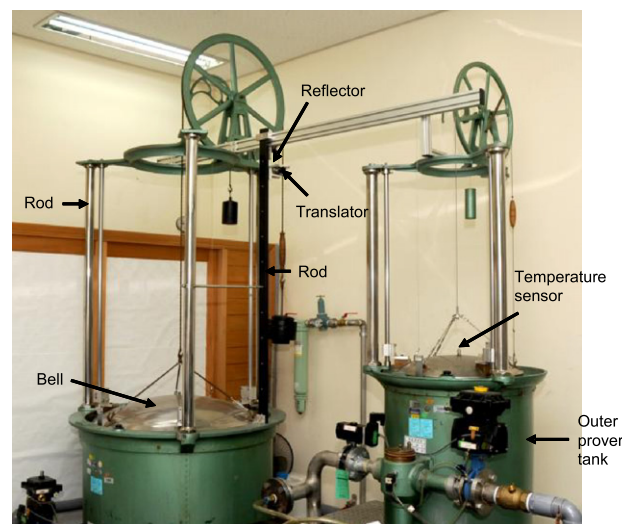


Fig. 1. Bell provers.

the center of the bell for accurate temperature or pressure measurement inside the bell. Thus, it was required to improve the system quality and perform a reanalysis of the uncertainty of the bell provers, shown in Fig. 1.

The first improvement was to install a laser interferometer. The moving distance of bell was measured by means of the laser interferometer, and the data collection time was automatically measured by connecting the timer to the laser interferometer. Also, to prevent the rocking of the bell and ensure the vertical rising of the bell, vertical bars (diameter 25 mm) were installed on three points of the outer prover tank, and these bars and the bell were connected and fastened with bearings. The three-way turn buckle above the bell was finely adjusted horizontally so that the bell is perpendicular. The bell will have a constant internal pressure during measurement regardless of flow rates, and will rise vertically without any rocking as this bar and the turn buckle are adjusted. The temperature measurement point was moved from the entrance of the bell prover to the inside of it.

In order to participate in intercomparison experiments as a standard system or to calibrate gas flow meters from industry, the uncertainty of the standard system must be evaluated. The standards organizations in many countries are already analyzing the accuracy of their systems, and are conducting tests on the characteristics of gas flows in accordance with test methods recommended in the international standards [2,3]. However, in previous studies of gas flow metrology, the measurement uncertainty is only explained briefly and it is hard to find any studies that have evaluated uncertainty in detail except for the sonic nozzle and the piston prover [4–7]. Ishibashi and Morioka [8] and Wright et al. [9] present an uncertainty calculation results for PVTt (Pressure, Volume, Temperature and time) systems which have a constant volume tank. Those contain valuable information on the measurement sensors, such as a detailed uncertainty analysis for the measurement of temperature and pressure.

Therefore the bell prover must have traceability in accordance with the national standards. And it must be able to test the flow rate of a gas meter and it is designed and produced to achieve automatic testing of a flow meter. Also, the bell prover has a stable time and distance measurement capability for the better performance of flow rate measurement and uncertainty which was evaluated in accordance with the ISO guide 25 [2].

2. Description of gas flow facility

The gas flow standard system is composed of an air compressor, storage tanks, air filters, air dryers, regulators for pressure control,

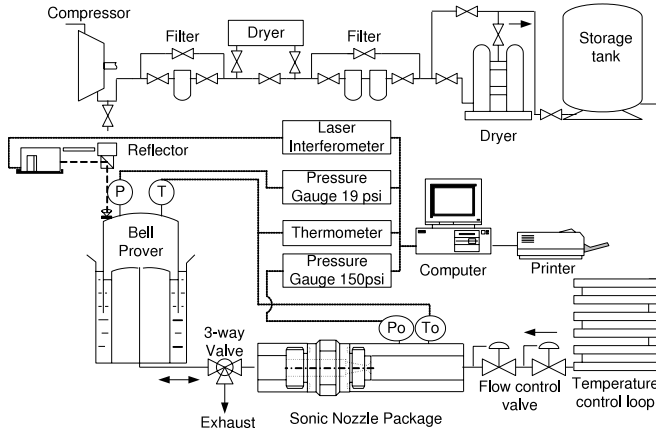


Fig. 2. Schematic diagram of a low-pressure gas flow standard system.

flow control valves, a flow bypass, platinum resistance thermometers, pressure gages, and a timer, as shown in Fig. 2. In the bell prover, two bell volumes (small: 0.2 m³, large: 0.6 m³) are used to measure the gas flow rate with the laser interferometer. The timing device is triggered by the signal generated as the reflector of the laser interferometer passes through designated points.

The compressor (37.5 kW, 288 m³/h) consists of an air-cooled system, air filters and oil filters. The compressed air, which is high in humidity and at a high temperature of about 311.15 K, passes through the air dryers and is then cooled below the 277.15 K saturation point. Condensed water is drained outside. The humidity and dust of the compressed air are filtered through the air filters. The capacity of treating the saturated air is 390 N m³/h. Dust, minute oil particles, and moisture above 5 μm are removed with the air dryers. The air through the air dryer with 213.15 K saturation point is stored in a tank of capacity 6.5 m³. The maximum allowable pressure of the storage tank is 1 MPa. Ten pipes to stabilize the temperature of the compressed air, with 0.1 m diameter and 6 m length, are connected in parallel and installed in the test room. Two pressure control valves are used to obtain stable pressure and flow rate in the test line.

All the temperature and pressure data of the gas meter, bell provers and various signals of the atmospheric pressure and temperature are collected by a data acquisition unit (HP34970A). The collecting time of the bell provers is measured with a timer (HP53131A). The two thermometers used in this facility are four-wire (100 Ω) platinum resistance thermometers. The temperature sensors measure the inlet gas temperature every 1 s. The pressures for the flow meter and the provers are measured by the differential pressure gages. All instruments are interfaced to the personal computer. Selection among the two bells and the data reduction are conducted in a Visual Basic program. The bell prover is installed in a temperature-controlled room at 294.15 ± 1 K.

3. Uncertainty evaluation

3.1. The mathematical model

A bell prover system consists of the pipes that connect every device, a switching valve, and standard volume bells. A “control volume” inside the pipes can be defined from the outlet of the test meter to the beginning of the standard volume bell that is in the bell prover. Conservation of mass produces the following formula:

$$0 = \frac{\partial}{\partial t} \int_V \rho \cdot dV + \int_A \rho \vec{v} \cdot d\vec{A}. \quad (1)$$

The mass flow rate q_m is given by

$$\begin{aligned} q_m &= - \int_A \rho \vec{v} \cdot d\vec{A} = \frac{\partial}{\partial t} \int_V \rho \cdot dV = \frac{m_e - m_b}{t} \\ &= \frac{1}{t} (V_c \rho_e + V_L \rho_e - V_L \rho_b) + m_\ell \\ &= \frac{1}{t} (V_c \rho_e + V_L \Delta \rho_{eb}) + m_\ell. \end{aligned} \quad (2)$$

After the air temperature inside the storage and bell is stabilized, the pressure change should be less than 2 Pa during 3 min. This means that m_ℓ is negligible during the calibration process.

The uncertainty of the mass flow rate, q_m of Eq. (2), yields

$$\begin{aligned} u(q_m) &= [C_{V_c}^2 u^2(V_c) + C_{\rho_e}^2 u^2(\rho_e) + C_{V_L}^2 u^2(V_L) \\ &\quad + C_{\Delta \rho_{eb}}^2 u^2(\Delta \rho_{eb}) + C_t^2 u^2(t)]^{\frac{1}{2}}. \end{aligned} \quad (3)$$

The sensitivity coefficients can be obtained by differentiation of Eq. (2); thus the relative sensitivity coefficients are, omitting units,

$$\begin{aligned} C_{V_c} &= \frac{\rho_e}{t}, \quad C_{\rho_e} = \frac{V_c}{t}, \quad C_{V_L} = \frac{\Delta \rho_{eb}}{t}, \\ C_{\Delta \rho_{eb}} &= \frac{V_L}{t}, \quad C_t = \frac{-1}{t^2} (V_c \rho_e + V_L \cdot \Delta \rho_{eb}). \end{aligned} \quad (4)$$

The relative uncertainty of the mass flow rate is expressed as follows:

$$\frac{u(q_m)}{q_m} = \left[\frac{1}{\frac{1}{t} (V_c \rho_e + V_L \Delta \rho_{eb})} u^2(q_m) \right]^{\frac{1}{2}} \cong \left[\frac{1}{\frac{1}{t} (V_c \rho_e)} u^2(q_m) \right]^{\frac{1}{2}}. \quad (5)$$

Eq. (5) with Eqs. (3) and (4) is reduced to

$$\begin{aligned} \frac{u(q_m)}{q_m} &= \left[\left[\frac{u(V_c)}{V_c} \right]^2 + \left[\frac{u(\rho_e)}{\rho_e} \right]^2 + \left[\frac{V_L}{V_c} \frac{u(\Delta \rho_{eb})}{\rho_e} \right]^2 \right. \\ &\quad \left. + \left[\frac{\Delta \rho_{eb}}{\rho_e} \frac{u(V_L)}{V_c} \right]^2 + \left[\frac{-u(t)}{t} \right]^2 \right]^{\frac{1}{2}}. \end{aligned} \quad (6)$$

3.2. Component of uncertainty evaluation

3.2.1. Uncertainty of collection volume $\left(\frac{u(V_c)}{V_c} \right)$

The collection volume (V_c) is expressed as follows [10]:

$$V_c = V_{out} - V_{oil,\ell} - V_{oil,ad}.$$

The uncertainty of collection volume is expressed as Eq. (7):

$$\frac{u(V_c)}{V_c} \cong \left[\left[\frac{u(V_{out})}{V_c} \right]^2 + \left[\frac{u(V_{oil,\ell})}{V_c} \right]^2 + \left[\frac{u(V_{oil,ad})}{V_c} \right]^2 \right]^{\frac{1}{2}}. \quad (7)$$

1) Uncertainty of volume outside of the bell $\left(\frac{u(V_{out})}{V_c} \right)$

The volume outside of the bell

$$V_{out} = \frac{\pi}{4} \cdot D_{out}^2 \cdot L.$$

The uncertainty of the volume outside of the bell is expressed as Eq. (8).

$$\frac{u(V_{out})}{V_c} = \left[\left[\frac{2u(D_{out})}{D_{out}} \right]^2 + \left[\frac{u(L)}{L} \right]^2 \right]^{\frac{1}{2}}. \quad (8)$$

(1) Measurement uncertainty of bell outside diameter $\left(\frac{u(D_{out})}{D_{out}} \right)$

The bell outside diameters are measured with a standard π -tape in the Length Standard Center of the KRISS. The diameters are measured at 14 sections of the small bell prover and 10 sections of the large bell prover. Table 1 shows the uncertainty of the outside diameter of the small bell prover. The maximum uncertainty is

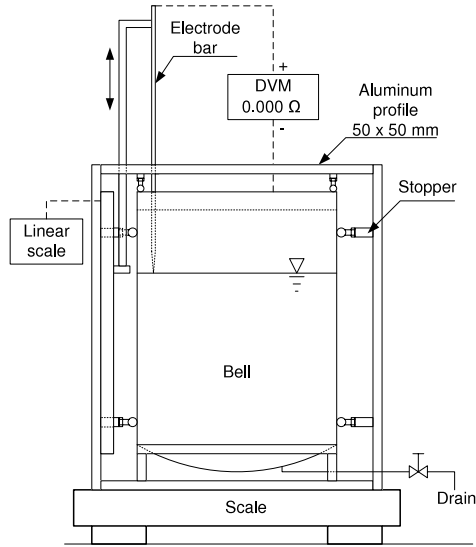


Fig. 3. Schematic diagram of the water draw method.

$5.9 \times 10^{-3}\%$ at 350 mm above the bell bottom and the maximum uncertainty of the large bell prover is $5.8 \times 10^{-3}\%$ at 260 mm above the bell bottom. The number of effective degrees of freedom (ν_{eff}) is calculated as shown in Eq. (9) using the Welch–Satterthwaite formula. The effective degrees of freedom number is 59.

$$\nu_{\text{eff}} = \frac{u_c^4(y_i)}{\sum_{i=1}^N \frac{[c_i u(x_i)]^4}{\nu_i}} = \frac{(5.9 \times 10^{-3})^4}{\frac{(1.3 \times 10^{-3})^4}{6} + \frac{(5.8 \times 10^{-3})^4}{55}} \approx 59. \quad (9)$$

The π -tape measured the outside diameter based on the assumption of a circular cross section and constant thickness of the bell. However, a slightly elliptical bias of a circular cross section might lead to an additional error. To directly measure the volume of the bell, the KRISS used the water draw method, i.e. directly replacing the volume of the bell with water, as illustrated in Fig. 3. This method measures the weight of the bell to calculate the volume. And the bell inside diameter can also be calculated from the volume of water and extracted water height. Thus, the bell inside diameter obtained by the water draw method can be compared with the value obtained by using the π -tape and the uncertainty of the roundness of the bell inside diameter can be evaluated.

The frame is made of an aluminum profile (0.05×0.05 m) to erect the bell vertically and fasten it to the scale for measuring the weight of the bell (resolution 1 g), and a linear scale (resolution $5 \mu\text{m}$) was attached to the frame to measure the water height. To measure the water height, an electrode bar was installed as illustrated in Fig. 3, and the two electrodes were used to measure the height on the basis of the theory of electric currents flowing through water. The ends of the electrodes were made as sharp and long as possible to eliminate the uncertainty of height measurement due to the influence of the surface tension of water. The inside of the bell was filled with water. After the temperature of the water was equal to room temperature, the height, weight and temperature of the water were measured. And the atmospheric pressure was also measured while draining the water at an interval of height 80 mm. The density of water was calculated to verify the volume and the bell inside diameter.

Table 2 shows the bell outside diameter measurements obtained by the water draw method and the π -tape method. As a result of Table 2, the maximum deviation of the outside diameter between the π -tape and water draw methods is 0.27 mm, and the uncertainty due to elliptical bias was $5.1 \times 10^{-5}\%$ [11]. As the water draw method has many factors contributing to measurement

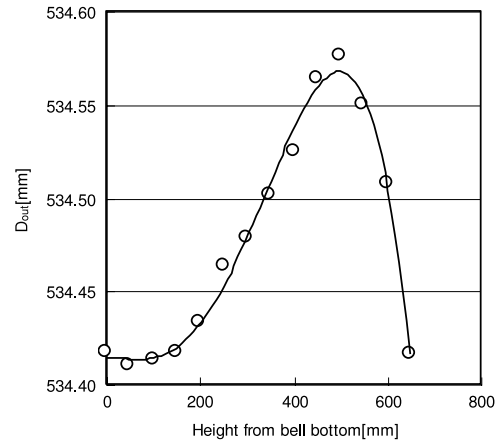


Fig. 4. Calibration curve for the diameter of the small bell prover.

uncertainty, such as the calibration of the length scale, the thermometer, pressure gauge and height measuring instrument which is using an electrode based on the surface tension of water, and the vertical installation of the bell, it is difficult to accurately measure the bell inside diameter in the field according to the water draw method. Furthermore, the diameter of the large bell (0.6 m^3) is not measured by the water draw method because of the possibility of the bell deformation or cracks on the welding parts caused by the mass of water in the bell. Consequently, in this study the values measured according to the π -tape method were used to evaluate the uncertainty.

To reduce the uncertainty due to the non-uniformity of the bell area, the measured diameter was programmed in the computer with a curve fitting equation, as illustrated in Fig. 4. If this curve fitting is used, the maximum deviation of the diameter is 0.15 mm. Therefore the uncertainty due to non-uniformity of the bell diameters is maximum $2.4 \times 10^{-2}\%$ for the two bell provers.

The temperature of the laboratory where the bell prover is located was $294.15 \pm 1.0 \text{ K}$. The material of bell is stainless steel, and the thermal expansion coefficient is $14.7 \times 10^{-6} / \text{K}$ [12]. Standard uncertainties in diameter due to thermal expansion are $7.9 \times 10^{-5} \text{ mm}$ for the small bell prover and $1.3 \times 10^{-4} \text{ mm}$ for the large bell prover. The relative standard uncertainty of thermal expansion is $2.0 \times 10^{-5}\%$, which is negligible compared with other factors.

The uncertainties of the bell outside diameter are given in Table 3. The number of effective degrees of freedom is infinite.

The uncertainty of the bell outside diameter is

$$\frac{u(D_{\text{out}})}{D_{\text{out}}} = 2.5 \times 10^{-2}\% \quad (\nu_{\text{eff}} = \infty).$$

(2) Uncertainty of bell stroke distance $\left(\frac{u(L)}{L}\right)$

The stroke length of the two bells is measured with a laser interferometer. The calibration of the laser interferometer is not required for this measurement purpose because of the frequency stability of the laser light source. The resolution in this measurement is $0.1 \mu\text{m}$. The uncertainty of laser interferometer resolution is $5.0 \times 10^{-4}\%$ with a rectangular probability distribution, the standard uncertainty estimation is $0.1 \mu\text{m} / 2\sqrt{3} = 0.029 \mu\text{m}$, and the number of degrees of freedom is infinite.

Another error source is the vertical alignment of the axial direction of the bell to the laser light. The laser light from the laser interferometer arrives at the reflector to measure the traveling height of the bell through the mirror, and the light is then reflected and returns to the laser interferometer. To install the reflector, a precise level indicator can be used for leveling, and the x–y translator can be used so that the laser is focused on the center of

Table 1
Outside diameter.

Height from bell bottom (mm)	Average diameter (mm)	$u_A \times 10^{-3}\%$	$u_B \times 10^{-3}\%$	$u_C \times 10^{-3}\%$	v_{eff}
100	534.417	0.7		5.8	56
150	534.410	0.6		5.8	56
200	534.413	0.5		5.8	56
250	534.417	0.8		5.8	57
300	534.433	0.7		5.8	56
350	534.463	1.3		5.9	59
400	534.479	0.8		5.8	57
450	534.501	0.6	5.8	5.8	56
500	534.524	1.0		5.9	58
550	534.564	0.6		5.8	56
600	534.576	0.6		5.8	56
650	534.550	0.9		5.8	57
700	534.507	1.1		5.9	58
750	534.416	0.4		5.8	58

Table 2
Results of bell outside diameter measurements (small bell prover).

Height from bell bottom (mm)	π -tape method (mm)	Water draw method (mm)	Difference of diameter (mm)
82.6	534.420	534.693	0.27
164.1	534.449	534.655	0.21
245.9	534.495	534.516	0.02
328.5	534.543	534.457	0.09
412.3	534.563	534.585	0.02
495.4	534.520	534.543	0.02
576.8	534.367	534.464	0.10

Table 3
Uncertainties of bell outside diameter.

Prover	π -tape (%)	Elliptical bias (%)	Non-uniformity (%)	$u(D_{\text{out}})/D_{\text{out}}$ (%)	v_{eff}
Small	5.9×10^{-3}	5.1×10^{-5}	2.4×10^{-2}	2.5×10^{-2}	∞
Large	5.8×10^{-3}			2.5×10^{-2}	∞

Table 4
Uncertainties for the volume outside of the bell, $\left(\frac{u(V_{\text{out}})}{V_c}\right)$.

Prover	$u(D_{\text{out}})/D_{\text{out}}$ (%)	$u(L)/L$ (%)	$u(V_{\text{out}})/V_c$ (%)	v_{eff}
Small	2.5×10^{-2}	5.0×10^{-4}	2.5×10^{-2}	∞
Large	2.5×10^{-2}		2.5×10^{-2}	∞

Table 5
Uncertainties of volume outside of the bell due to the oil level.

Prover	$V_{\text{oil},\ell}/V_c$ (%)	$u(V_{\text{oil},\ell})/V_{\text{oil},\ell}$ (%)	$u(V_{\text{oil},\ell})/V_c$ (%)	v_{eff}
Small	0.26	1.9×10^{-3}	5.0×10^{-4}	6
Large	0.30	7.0×10^{-4}	2.0×10^{-4}	6

the reflector. The reflector was installed on the bar (1.4×0.46 m) with a bearing. For the measurement of the vertical alignment of the axial direction of the bell, a micrometer (resolution $10 \mu\text{m}$) on the translator of the reflector was used to test whether the laser interferometer detected the light from the reflector. The measurement was done by moving the reflector from the center of the laser light by 0.1 mm. With a misalignment of 0.5 mm, the laser interferometer was unable to detect the laser light from the reflector. Therefore, the maximum vertical misalignment of the axial direction is 0.5 mm from the center of the reflector. The stroking distance of the small bell prover is 620 mm. The angular error has to be less than 0.05° . So the uncertainty of the vertical stroking distance measurement of the bell is $(1 - \cos(0.05^\circ)) \times 100 = 38 \times 10^{-5}\%$. Taking this value as the half range with a rectangular probability distribution, the standard uncertainty is $22 \times 10^{-5}\%$.

The two standard uncertainties of stroke distance were combined by the root sum square method to give a result of $5.0 \times 10^{-4}\%$.

Therefore, the uncertainties for the bell outside volume are as given in Table 4. And the number of effective degrees of freedom is infinite for small and large provers.

$$\frac{u(V_{\text{out}})}{V_c} = 2.5 \times 10^{-2}\% \quad (v_{\text{eff}} = \infty).$$

2) Uncertainty of volume outside of the bell due to the oil level

The uncertainty of the oil level must be evaluated because the oil level changes due to the pressure changing inside the bell, which is influenced to the volume of the bell when the bell is rising up. The oil level change due to the bell displacement is (5.329 ± 0.060) mm for the small bell prover and (14.380 ± 0.059) mm for the large bell prover. The oil level was measured with a vernier caliper. Changing of oil level is caused by the pressure change in the bell, and another reason is the rising up the submerged wall of the bell from the oil. The amount of correction due to oil displacement is calculated from the total volume change of oil divided by the area between the bell outside surface and outer prover tank. The uncertainty of the volume outside the bell due to the oil level changing is given in Table 5. The number of effective degrees of freedom is 6.

$$\frac{u(V_{\text{oil},\ell})}{V_c} = 5.0 \times 10^{-4}\% \quad (v_{\text{eff}} = 6).$$

3) Uncertainty of volume due to oil film adherence $\left(\frac{u(V_{\text{oil},\text{ad}})}{V_c}\right)$

When the bell is raised from the oil bath, part of the oil is adhering to the bell, which reduces the collection volume inside the bell. The volume of oil adhering to the bell, $V_{\text{oil},\text{ad}}$, is given as Eq. (10) [11].

$$V_{\text{oil},\text{ad}} = \frac{2 \cdot \pi}{3} \left(\frac{v \cdot v_B}{g} \right)^{\frac{1}{2}} \cdot L \cdot D_{\text{in}}. \quad (10)$$

Table 6
Uncertainty due to oil film adherence.

Prover	Units	Small	Large
v_B	cm/s	1.17	3.41
v	mm ² /s	3.392	3.392
L	cm	62	100
D_{in}	cm	53.316	86.791
$V_{oil,ad}$	cm ³	44.034	197.381
$u(V_{oil,ad})/V_c$	%	0.02	0.02
v_{eff}		∞	∞

Table 7Uncertainty of the collection volume, $\left(\frac{u(V_c)}{V_c}\right)$.

Prover	$u(V_{out})$ (%)	$u(V_{oil,t})$ (%)	$u(V_{oil,ad})$ (%)	$u(V_c)/V_c$ (%)	v_{eff}
Small	2.5×10^{-2}	5.0×10^{-4}	2.0×10^{-2}	3.2×10^{-2}	∞
Large	2.5×10^{-2}	2.0×10^{-4}	2.0×10^{-2}	3.2×10^{-2}	∞

The uncertainty results of Eq. (10) for the two bell provers are given in Table 6. The relative standard uncertainty is $2.0 \times 10^{-2}\%$ based on a rectangular distribution. The number of degrees of freedom is infinite.

Substituting the above three uncertainty items into Eq. (7) gives the relative standard uncertainty, as shown in Table 7.

$$\frac{u(V_c)}{V_c} = 3.2 \times 10^{-2}\% \quad (v_{eff} = \infty).$$

3.2.2. Uncertainty of the air density measurement $\left(\frac{u(\rho_e)}{\rho_e}\right)$

The air density is expressed as

$$\rho_e = \frac{p_e}{Z(p_e, T_e)RT_e}.$$

The relative uncertainty of the air density is reduced to

$$\frac{u(\rho_e)}{\rho_e} = \left[\left[\frac{u(p_e)}{p_e} \right]^2 + \left[\frac{u(T_e)}{T_e} \right]^2 + \left[\frac{u(RZ(p_e, T_e))}{RZ(p_e, T_e)} \right]^2 \right]^{\frac{1}{2}}. \quad (11)$$

1) Uncertainty of the pressure measurement $\left(\frac{u(p_e)}{p_e}\right)$

The uncertainties of the barometer and the differential pressure gauge (type B) are $5.0 \times 10^{-3}\%$ and $3.5 \times 10^{-2}\%$ of full scales, respectively. The number of degrees of freedom is 41 for the barometer and 9 for the differential pressure gauge. The type A and B uncertainties and the relative standard uncertainty of the pressure measurement are given in Table 8. The number of effective degrees of freedom is 7.

2) Uncertainty of the temperature measurement $\left(\frac{u(T_e)}{T_e}\right)$

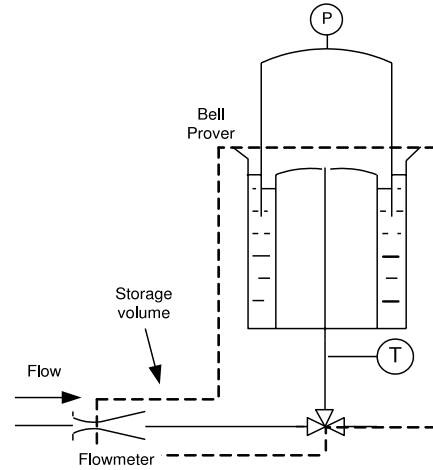
The temperature sensor is calibrated with the data acquisition system and the calibration curve is used. The type B uncertainty of the temperature measurement is $1.7 \times 10^{-2}\%$. The type A and B uncertainties and the relative standard uncertainty of the temperature measurement are given in Table 8. The number of effective degrees of freedom is 54.

3) Uncertainties of gas constant, R , and compressibility, Z

The uncertainty of the gas constant R is zero because it is constant and the uncertainty of the gas compressibility constant Z derived from NASA TN D-2565 [13] is $2.0 \times 10^{-2}\%$, while the number of degrees of freedom is infinite.

The relative standard uncertainty of the air density measurement is shown in Table 8. And the number of effective degrees of freedom is 17.

$$\frac{u(t)}{t} = 4.4 \times 10^{-2}\% \quad (v_{eff} = 17).$$

**Fig. 5.** Storage volume.

3.2.3. Uncertainty of the mass change in the storage volume $\left(\frac{\Delta m_L}{m_c}\right)$

The pressure and temperature differences in the storage volume during the calibration process are given in Fig. 5. The ratio of mass change in the storage volume to the mass of bell is expressed as

$$\begin{aligned} \frac{\Delta m_L}{m_c} &= \frac{1}{V_c \rho_e} [V_L (\rho_e - \rho_b)] \\ &= \frac{V_L}{V_c} \left(1 - \frac{\rho_b}{\rho_e} \right) \\ &= \frac{V_L}{V_c} \left(1 - \frac{p_b}{p_e} \cdot \frac{T_e}{T_b} \right) \\ &= \frac{V_L}{V_c} \left[1 - \left(1 + \frac{\Delta T}{T_b} \right) / \left(1 + \frac{\Delta P}{P_b} \right) \right]. \end{aligned} \quad (12)$$

The calculated results using Eq. (12) for the three flow rates (low (L), middle (M), and high (H)) in each bell are given in Table 9. The value of $\frac{\Delta m_L}{m_c}$ is larger than the sum of the uncertainties introduced by the storage volume $\frac{V_L}{V_c} \frac{u(\Delta \rho_{eb})}{\rho_e}$ and $\frac{\Delta \rho_{eb}}{\rho_e} \frac{u(V_L)}{V_c}$ in Eq. (6). And the value of $\frac{\Delta m_L}{m_c}$ in Table 9 may not affect the expanded uncertainty. So the sum of $\frac{V_L}{V_c} \frac{u(\Delta \rho_{eb})}{\rho_e}$ and $\frac{\Delta \rho_{eb}}{\rho_e} \frac{u(V_L)}{V_c}$ is assumed to be $\frac{V_L}{V_c} \left[1 - \left(1 + \frac{\Delta T}{T_b} \right) / \left(1 + \frac{\Delta P}{P_b} \right) \right]$ in this study. The number of degrees of freedom is infinite.

3.2.4. Uncertainty due to the time measurement $\left(\frac{u(t)}{t}\right)$

The timer (HP53131A) is used to measure the diversion time. The uncertainty of the time measurement is divided into the uncertainties of the timer itself and the timing error produced by switching the valves.

The uncertainty of time measurement is expressed as Eq. (13).

$$\frac{u(t)}{t} = \left[\left[\frac{u(t_m)}{t} \right]^2 + \left[\frac{u(t_R)}{t} \right]^2 + \left[\frac{u(t_V)}{t} \right]^2 + \left[\frac{u(t_{BR})}{t} \right]^2 \right]^{\frac{1}{2}}. \quad (13)$$

1) Timer (HP5316A) $\left(\frac{u(t_m)}{t}\right)$

The bias of the timer at 10 MHz is 1.2×10^{-2} Hz and the uncertainty is 6.0×10^{-5} Hz ($k = 2$) at the 95% confidence level. The bias of the timer can be used as the uncertainty of the timer and the uncertainty of timer can be neglected because it is very small value compared with the bias. The air collection time of the bell prover is at least 40 s, so the uncertainty of the timer is

$$1.2 \times 10^{-2} \text{ Hz} / (1.0 \times 10^7 \text{ Hz}) 40 \text{ s} = 4.8 \times 10^{-8} \text{ s}.$$

Table 8Uncertainty of the air density measurement $\left(\frac{u(\rho_e)}{\rho_e}\right)$.

Prover	$u(P_e)/P_e$		$u(T_e)/T_e$		$u(z)/z$ (%)	$u(\rho_e)/\rho_e$ (%)	ν_{eff}
	u_A (%)	u_B (%)	u_A (%)	u_B (%)			
Small	2×10^{-3}	3.5×10^{-2}	2×10^{-3}	1.7×10^{-2}	2.0×10^{-2}	4.4×10^{-2}	17
Large	2×10^{-3}	3.5×10^{-2}	4×10^{-3}	1.7×10^{-2}	2.0×10^{-2}	4.4×10^{-2}	17

Table 9

Uncertainty of storage volume.

Prover	q_m	V_L (m ³)	V_c (m ³)	$\Delta T/T_b$	$\Delta P/P_b$	$\Delta m_L/m_c$ (%)
Small	L	4.5×10^{-2}	0.13	-1.6×10^{-4}	7.4×10^{-6}	5.8×10^{-3}
	M			-6.8×10^{-5}	4.4×10^{-6}	2.5×10^{-3}
	H			-6.0×10^{-5}	1.4×10^{-5}	2.6×10^{-3}
Large	L	1.5×10^{-1}	0.57	9.5×10^{-5}	-2.0×10^{-5}	3.1×10^{-3}
	M			9.3×10^{-5}	2.2×10^{-5}	1.9×10^{-3}
	H			1.0×10^{-4}	-1.4×10^{-5}	3.1×10^{-3}

Therefore, the relative standard uncertainty of the pressure measurement is

$$4.8 \times 10^{-8} \text{ s}/40 \text{ s} \times 100 = 1.2 \times 10^{-7}\%.$$

The time measurement uncertainty is less than $1.2 \times 10^{-7}\%$ due to the bias and the uncertainty of the timer based on the collection time of 40 s.

2) Timer resolution $\left(\frac{u(t_R)}{t}\right)$

The resolution of the timer is 1.0×10^{-3} s and the probability is assumed to have a rectangular distribution. The relative standard uncertainty is $(1.0 \times 10^{-3} \text{ s}/2\sqrt{3})/40 \text{ s} \times 100 = 7.0 \times 10^{-4}\%$ based on the collection time of 40 s. The number of degrees of freedom is also infinite.

3) Timing error of the valve $\left(\frac{u(t_V)}{t}\right)$

The flow direction is changed by three-way solenoid valves. A three-way valve makes the flow rate change while it is being closed or opened. The collection time error calculation method of the three-way valve operation suggested by Olsen and Baumgarten [14] is given in Eq. (14). A sonic nozzle [15] is used as a reference flow meter to test the timing error of valve because it shows very accurate performance and reliable stability during experiments.

$$C_{dm} = C_d \left(1 - \frac{t_V}{t}\right). \quad (14)$$

Therefore one could get an ideal value of the discharge coefficient to extend the collection time to infinity. Using the small bell prover and changing the collection time from 10 to 160 s, a sonic nozzle was calibrated to obtain the discharge coefficients. The collection time was measured with a timer (HP 53131A) automatically to pre-set the traveling height of the bell. The traveling height of the bell is measured by the laser interferometer.

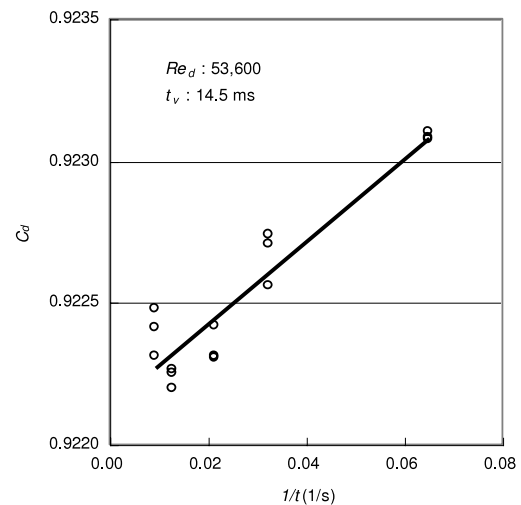
The test results are presented in the Fig. 6, and the correlation equation of the C_{dm} and t_V is as below.

$$C_{dm} = 0.9221 \left(1 + \frac{0.0145}{t}\right). \quad (15)$$

Thus the timing error of the air collection time is 1.45×10^{-2} s, and the relative standard uncertainty is $1.45 \times 10^{-2} \text{ s}/40 \text{ s} \times 100 = 36 \times 10^{-2}\%$ based on the minimum collection time 40 s.

4) Timing error of the bell rocking $\left(\frac{u(t_{BR})}{t}\right)$

The bell rocking effect can be considered as either a bell volume uncertainty or a time measurement uncertainty. Bell rocking is a serious problem which happens between the roller on the bell and the three guiding rods. That causes unstable movement of the

**Fig. 6.** Error of air collection time due to operation.

bell, fluctuation of pressure in the measuring volume and an error source of bell volume or time measurement. The main reason for the unstable movement of the bell is an unstable friction due to the contact between the roller on the bell and the three guiding rods. To solve this problem, the old rods and the rollers were replaced with precision rods and ball bearings to make sure that the bell rises vertically to eliminate the rocking of the bell movement. So the rocking effect which will be contained in the instability of pressure in the bell is not considered as an error source of uncertainty.

The combined relative standard uncertainty of the time measurement and the number of degrees of freedom are calculated using the values listed in Sections 1)–4).

$$\frac{u(t)}{t} = 3.6 \times 10^{-2}\% \quad (\nu_{\text{eff}} = 39).$$

3.3. Expanded uncertainty

The expanded uncertainty at the 95% confidence level is obtained by multiplying the combined standard uncertainty by the coverage factor corresponding to the effective number of degrees of freedom. Based on the effective number of degrees of freedom, the coverage factor k at the 95% confidence level found from the t -distribution is 2. As shown in Table 10, the larger value of the uncertainty budget is 0.13% with 95% confidence level, which is rather than 0.17% of previous one. The main change of uncertainty

Table 10
Uncertainty budget of bell provers.

Components of uncertainty	Small		Large	
	$c_i u(x_i)/x_i$ (%)	ν_{eff}	$c_i u(x_i)/x_i$ (%)	ν_{eff}
$u(V_c)$	3.2×10^{-2}	∞	3.2×10^{-2}	∞
$u(V_{\text{out}})$	2.5×10^{-2}		2.5×10^{-2}	
$u(V_{\text{oil,level}})$	5.0×10^{-4}		2.0×10^{-4}	
$u(V_{\text{oil,ad}})$	2.0×10^{-2}		2.0×10^{-2}	
$u(\rho_e)$	4.4×10^{-2}	17	4.4×10^{-2}	17
$u(P_{eA})$	2.0×10^{-3}		2.0×10^{-3}	
$u(P_{eB})$	3.5×10^{-2}		3.5×10^{-2}	
$u(T_{eA})$	2.0×10^{-3}		2.0×10^{-3}	
$u(T_{eB})$	1.7×10^{-2}		1.7×10^{-2}	
$u(z)$	2.0×10^{-2}		2.0×10^{-2}	
$\Delta m_L/m_c$	6.0×10^{-3}	∞	6.0×10^{-3}	∞
$u(t)$	3.6×10^{-2}	39	3.6×10^{-2}	39
$u(t_m)$	1.2×10^{-7}		1.2×10^{-7}	
$u(t_R)$	1.5×10^{-3}		1.5×10^{-3}	
$u(t_V)$	3.6×10^{-2}		3.6×10^{-2}	
u_c	6.5×10^{-2}	70	6.5×10^{-2}	70
k	2		2	
U	0.13		0.13	

Table 11
Comparison of old and new uncertainty budgets.

Components of uncertainty	Old (%)	New (%)
$u(V_{\text{out}})$	5.0×10^{-2}	2.5×10^{-2}
$u(P_{eA})$	1.2×10^{-2}	2.0×10^{-3}
$u(T_{eA})$	3.1×10^{-2}	2.0×10^{-3}

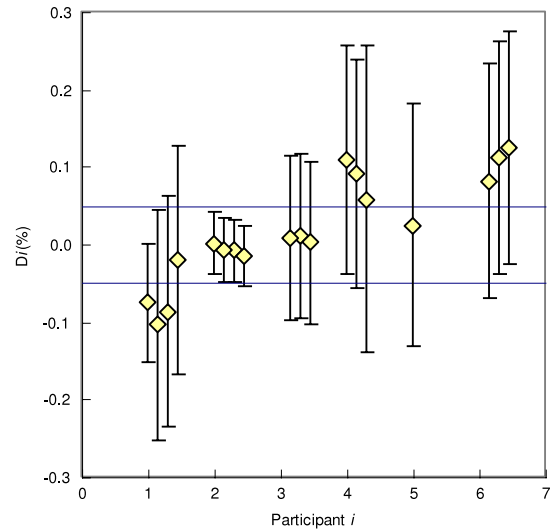
budget of the old and new systems is shown in Table 11. One of the main changes is because of the enhancement of length measurement accuracy by using a laser interferometer instead of the steel scale of the old system. Another improvement is pressure stability inside the bell using three precise guide rods with bearings. And the last improvement is the temperature stability inside the bell because of the relocation of the sensor position.

4. Intercomparison

The KRISS participated in a key comparison CCM.FF-K6 between six NMIs in the area of low-pressure gas flow which was organized by the CIPM (Comité International des Poids et Mesures) and the Working Group for Fluid Flow [16]. The main purpose of the comparison is to verify calibration and measurement capabilities by (1) determining key comparison reference values (KCRVs), (2) calculating the difference between each participant's results and the KCRV (D_i), and (3) calculating the degrees of equivalence between the participants, all with uncertainty estimates (the “KC outputs”). Also it is a good practice to check the effectiveness of the result of the bell prover uncertainty evaluation procedure of the KRISS. The participants, the types of primary standards used and the uncertainties of the facilities are listed in Table 12. The uncertainties given in Table 12 are standard ($k = 1$) values.

Table 12
Facilities used during the key comparison CCM.FF-K6.

Participant	Type of primary standard	Standard uncertainty (%)
NIST (United States)	34 L and 677 L PVTt	0.025
PTB (Germany)	CFV working standards	0.04
	Positive displacement working standard	0.059 to 0.076
KRISS (Korea)	Bell prover	0.065
NMIJ (Japan)	Gravimetric	0.05
	PVTt	0.075 to 0.1
NMIA (Australia)	Bell prover	0.08
CENAM (Mexico)	Bell prover	0.075

**Fig. 7.** Intercomparison results (1.PTB, 2.NIST, 3.KRISS, 4.NMIJ, 5.NMIA, 6.CENAM).

The uncertainty of the KRISS is around the average value of the uncertainties of the participating NMIs.

The transfer standard (TS) was a set of six critical flow venturis with pressure and temperature sensors. The TS was used at seven mass flows of dry air: 4.4, 5, 5.6, 90, 130, 180, 260 g/min. The two distinct flow ranges of the comparison were selected so that the participants could use two different primary standards and give a broader view of laboratory capabilities. All flow ranges have limitations, and only two laboratories calibrated at all four large TS flows. The KCRVs for each flow set point (and the other KC outputs) were calculated using the uncertainty weighted mean or median following the approach of Cox [17]. The degree of equivalence (D_i , difference from the KCRV) for each participant at each flow is shown in Fig. 7.

Flows of 90, 130, 180, and 260 g/min (left to right) which were tested by the bell prover are represented as individual points clustered for each participant (1.PTB, 2.NIST, 3.KRISS, 4.NMIJ, 5.NMIA, 6.CENAM). Horizontal lines represent the $k = 2$ uncertainty of the KCRV (0.05%). Error bars are the $k = 2$ uncertainty in the degree of equivalence of each laboratory to the KCRV. D_i of the KRISS is located close to the KCRV and is a similar value to that of NIST, which has a PVTt system. The final report of this key comparison is listed in the BIPM key comparison data base “Key and Supplementary Comparison: Appendix B (<http://kcdb.bipm.org/>)” as CCM.FF-K6 low pressure gas flow.

5. Conclusions

The bell prover is used as a standard gas flow measurement system under normal pressure; its performance as a national standard system needed to be improved and the uncertainty of the flow rate measurement by it needed to be evaluated. The pressure and temperature measurement location was changed to the center

of the bell to make more accurate measurements. The most important improvement in the weak points of the conventional bell prover made in this study is to measure the moving distance and traveling time of the bell with a laser interferometer. To reduce the rocking effect, three precision vertical bars and bearings were replaced to eliminate the unstable friction, and to ensure stable vertical rising of the bell. After making those improvements, the uncertainty of the bells calculated by the procedure in this study is 0.13% at the confidence level of 95%.

To validate the effectiveness of the procedure of bell prover uncertainty evaluation and to verify the calibration and measurement capabilities, the KRISS participated in a key comparison CCM.FF-K6 of CIPM. The uncertainty at the KRISS is around an average value of uncertainties of the participating NMI. The difference of the KRISS test value from the KCRV is within 0.02%.

Previous studies on bell provers contained only brief details of any uncertainty estimation. The details of the uncertainty components and procedures to measure those error components in this study will be used as a guide to evaluate the uncertainty of bell prover systems.

References

- [1] ISO/IEC: general requirement for the competence of testing and calibration laboratory. ISO 17025. 1999.
- [2] ISO: guide to the expression of uncertainty in measurement. ISO guide 25. 1993.
- [3] ISO: measurement of fluid flow-estimation of uncertainty of a flowrate measurement. ISO 5168. 2002.
- [4] Cha TS, Park K-A, Choi YM, Choi HM, Yoon BH. An experimental study on the discharge coefficients of small sonic nozzle. *J Fluid Mach* 2000;3(2):44–9.
- [5] Choi YM, Park K-A, Park SO. Interference effect between sonic nozzles. *Flow Meas Instrum* 1997;8(2):113–9.
- [6] Choi HM, Park K-A, Choi YM. Evaluation of flow measurement uncertainty of sonic nozzle. *Trans KSME B* 2001;25(12):1898–904.
- [7] Choi HM, Park K-A, Oh YK, Choi YM. Improvement and uncertainty evaluation of mercury sealed piston prover using laser interferometer. *Flow Meas Instrum* 2009;20(4–5):200–5.
- [8] Ishibashi M, Morioka T. The renewed airflow standard system in Japan for 5–1000m³/h. *Flow Meas Instrum* 2006;17:153–61.
- [9] Wright JD, Jonson AN, Moldover MR. Design and uncertainty analysis for a PVTt gas flow standard. *J Res Natl Inst Stand Technol* 2003;108(1):21–47.
- [10] Singer: instructions for installation, operation, maintenance and calibration of American bell-type provers manual. 1965.
- [11] Wright JD, Mattingly GE. NIST calibration services for gas flow meters; Piston prover and bell prover gas flow facilities. National Institute of Standards and Technology, NIST Special Publication 250-49; 1998.
- [12] Chronological scientific tables: national astronomical observatory. Maruzen Co. Ltd.; 1992. p. 471.
- [13] Johnson RC. Real-gas effects in critical flow through nozzles and tabulated thermodynamic properties. NASA TN D-2565. NASA Lewis Research Center; 1965.
- [14] Olson L, Baumgarten GP. In: Vamah WE, Wyland H, editors. Gas flow measurement by collection time and density in a control volume, flow; Its measurement and control in science and industry, vol. 1, part 3. ISA; 1971. p. 1287–95.
- [15] ISO 9300: measurement of gas flow by means of critical flow venturi nozzles. ISO. 2005.
- [16] Wright J, Mikan B, Paton R, Park KA, Nakao S, Chahine K, Arias R. CIPM key comparison for low-pressure gas flow: CCM.FF-K6. Final report. CIPM; 2007.
- [17] Cox MG. The evaluation of key comparison data. 2002;39:589–95.